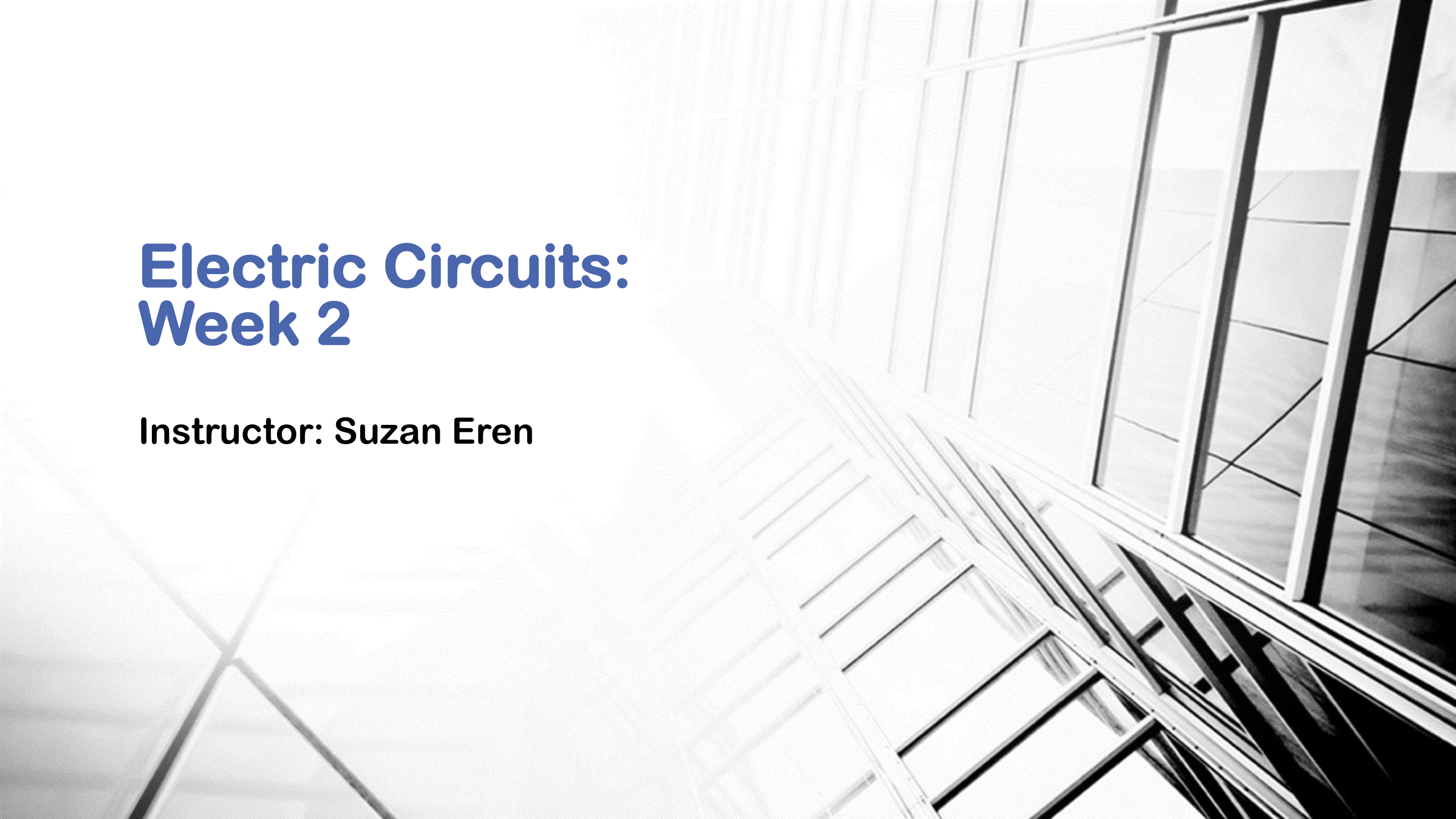
Electric Circuits: Week 2

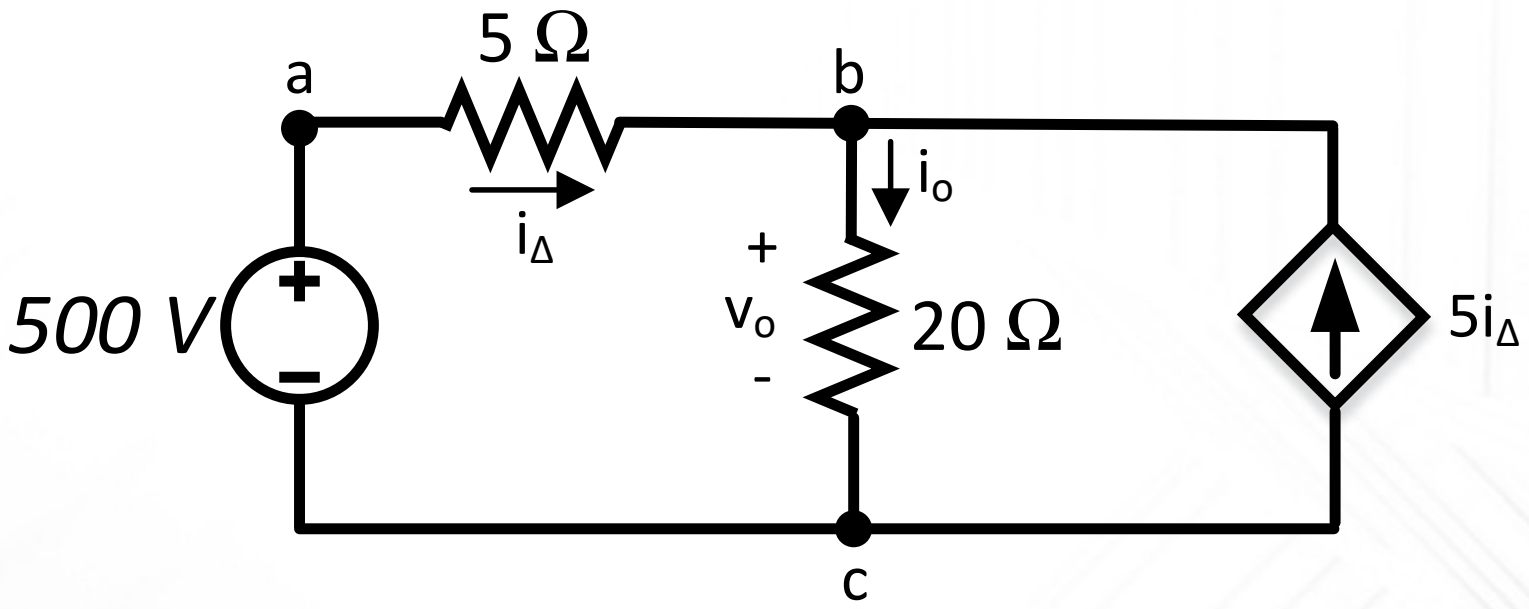
Instructor: Suzan Eren



Circuit Analysis (Continued…)

What happens when the circuit contains a dependent source?

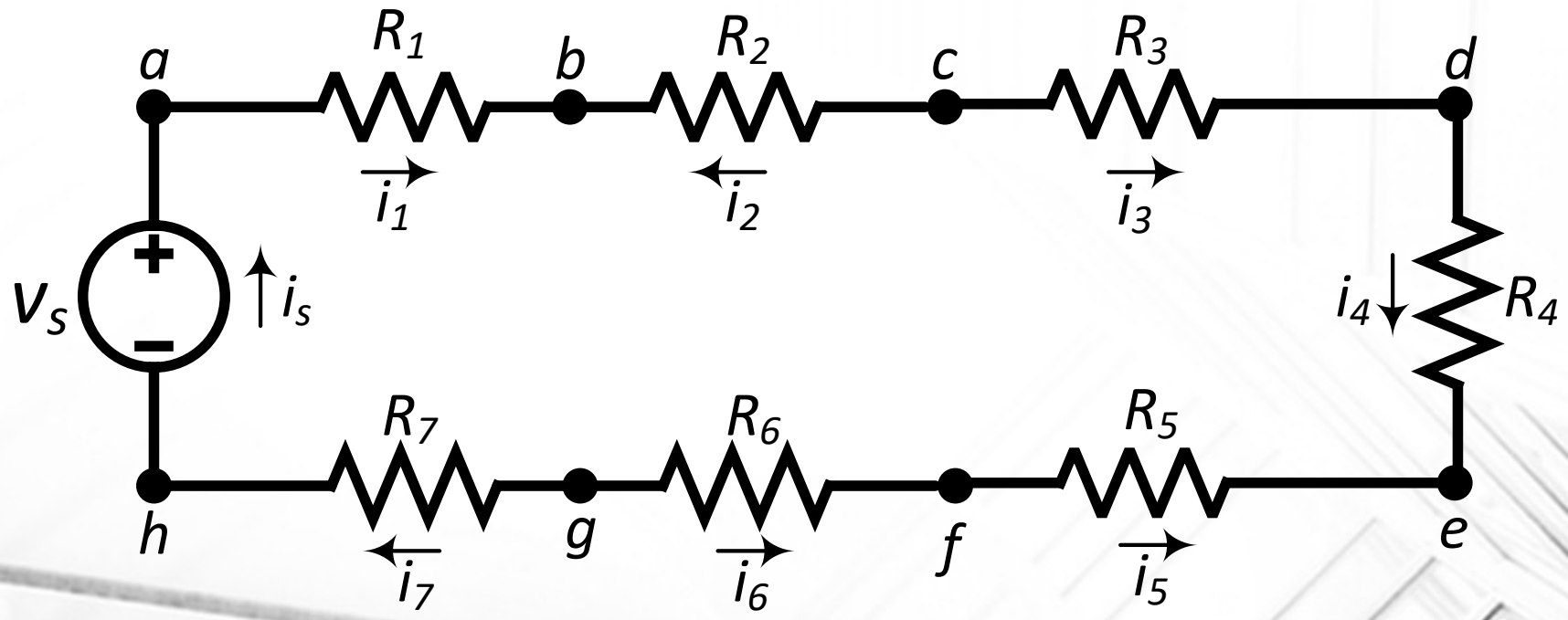
Example:



Resistors in Series

Refresher from Week 1: When there are only two elements connected to a single node, these two elements are said to be ***in series***.

Example: Which resistors are in series?



Equivalent Resistance

When resistors are in series, ***they can be combined to a single equivalent resistance.***

$$R_{eq} = \sum_{i=1}^{k} R_i = R_1 + R_2 + \ldots + R_k$$

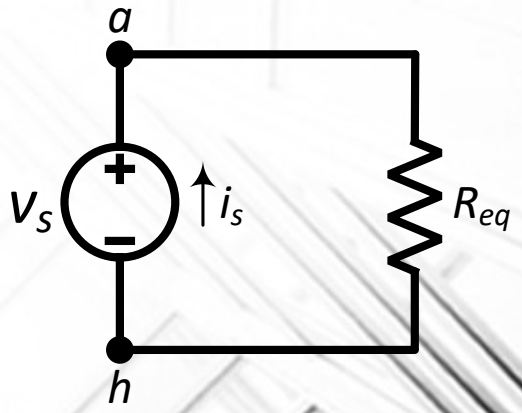
where k resistors are connected in series

Why? Current through them is the same. See previous example…

$$i_s = i_1 = -i_2 = i_3 = i_4 = -i_5 = -i_6 = i_7$$

Thus, we can rewrite the resistance as,

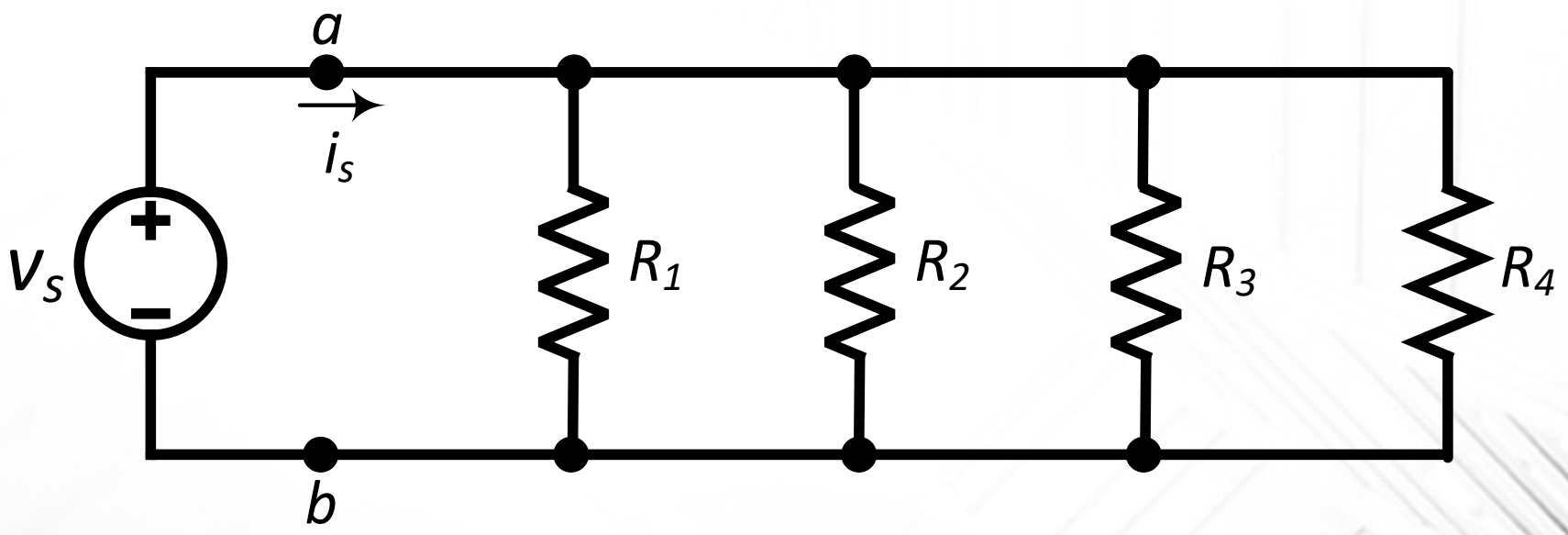
$$R_{eq} = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7$$



Resistors in Parallel

When two elements are connected at a single node pair, we can say that they ***in parallel***.

Example: Which resistors are in parallel?



Resistors in Parallel

When resistors are in parallel, ***they can be combined to a single equivalent resistance***.

$$\frac{1}{R_{eq}} = \sum_{i=1}^{k} \frac{1}{R_i} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

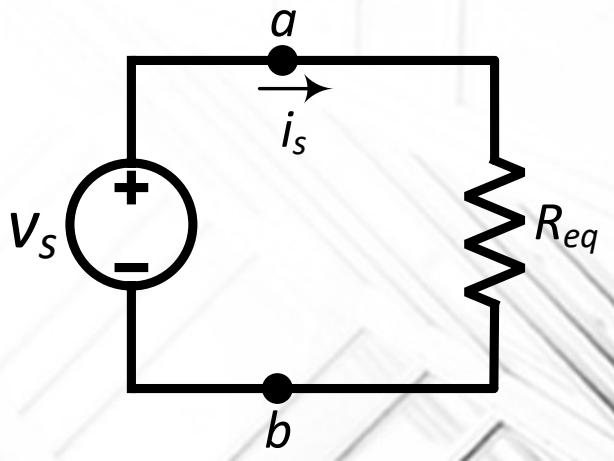
where k resistors are connected in parallel

Why? Voltage across them is the same. See previous example…

$$i_1 R_1 = i_2 R_2 = i_3 R_3 = i_4 R_4 = v_s$$

Thus, we can rewrite the resistance as,

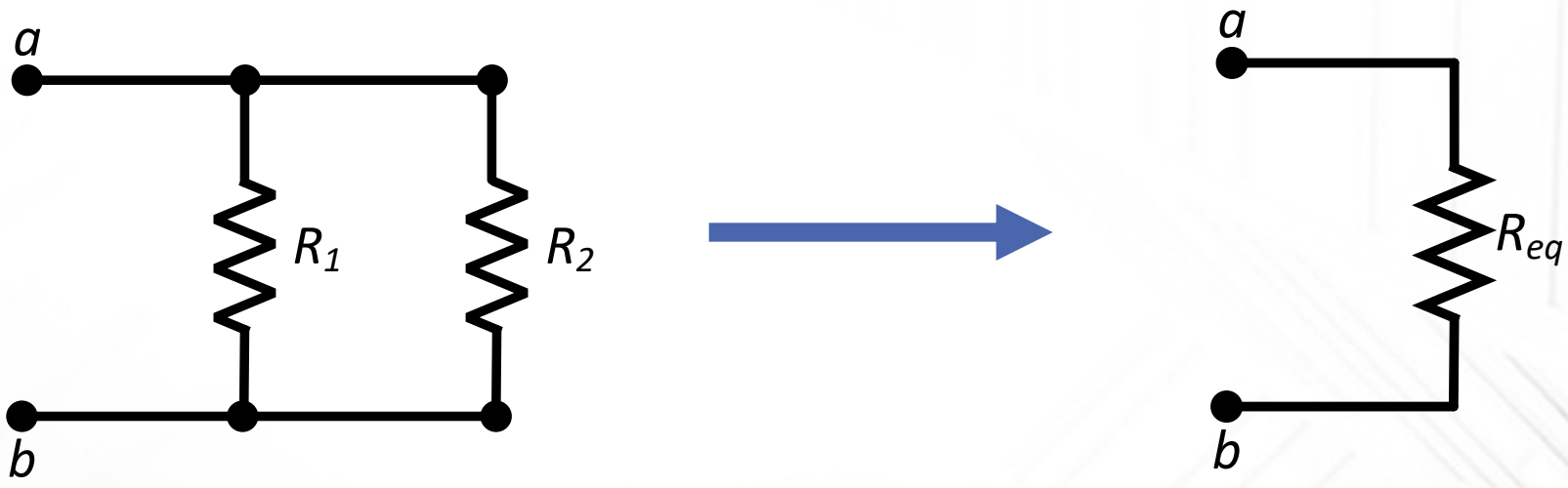
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$



Resistors in Parallel

When **two** resistors are in parallel, the equation can be simplified to:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$



Resistors in Series/Parallel

Interesting Note:

When resistors are in series:

$$R_{eq} > \max\left(R_1, R_2, \ldots, R_k\right)$$

When resistors are in parallel:

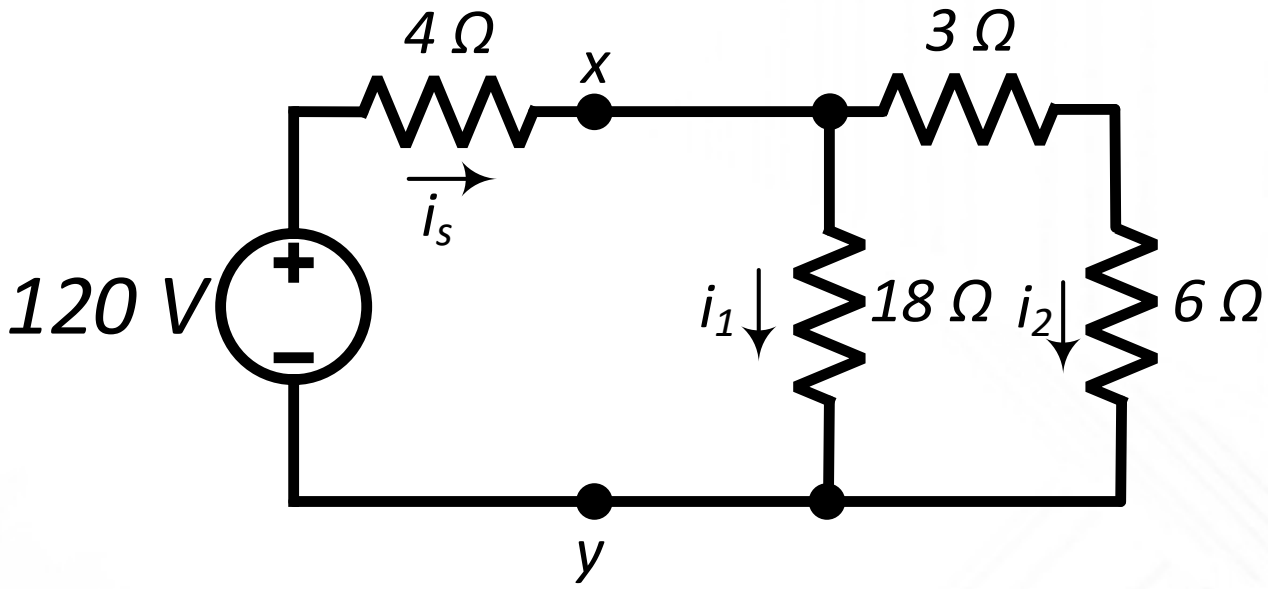
$$R_{eq} < \min\left(R_1, R_2, \ldots, R_k\right)$$

Also, we use this special notation for resistors in parallel: ||

Let's say we have two resistors in parallel, R_a and R_b. This can be rewritten as: $R_a || R_b$

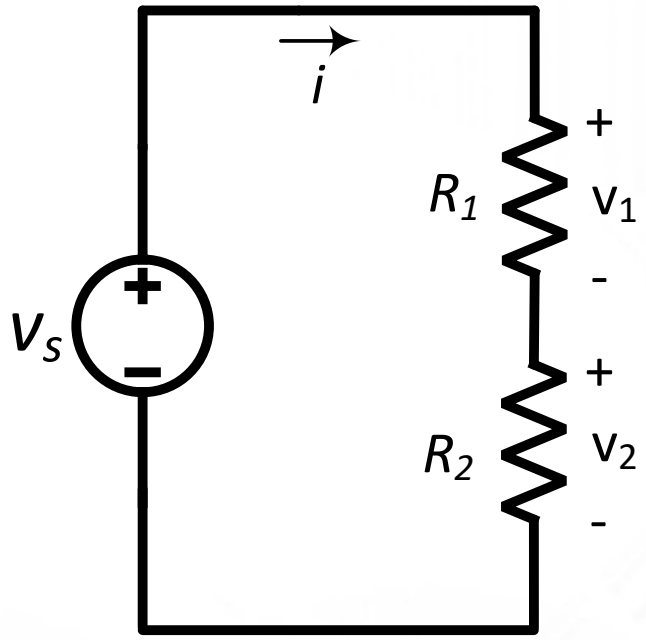
Example

Let's find the equivalent resistance, using the new information about resistors in series and in parallel…



Voltage Divider Circuit

When building an electric circuit, the number of voltage supplies at your disposal can be restricted. How do you get a different voltage level out of a single voltage supple? ***Voltage Dividers***

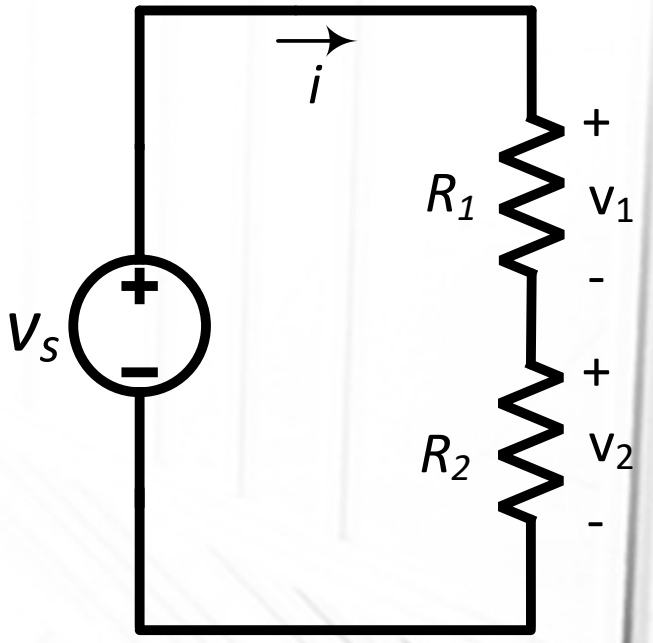


Voltage Divider Circuit

How do we figure out the different voltages across the two resistors?

KVL gives us: $v_s = iR_1 + iR_2$

which can be rewritten as: $i = \frac{v_s}{R_1+R_2}$



So, using Ohm's Law, we can find the two voltages…

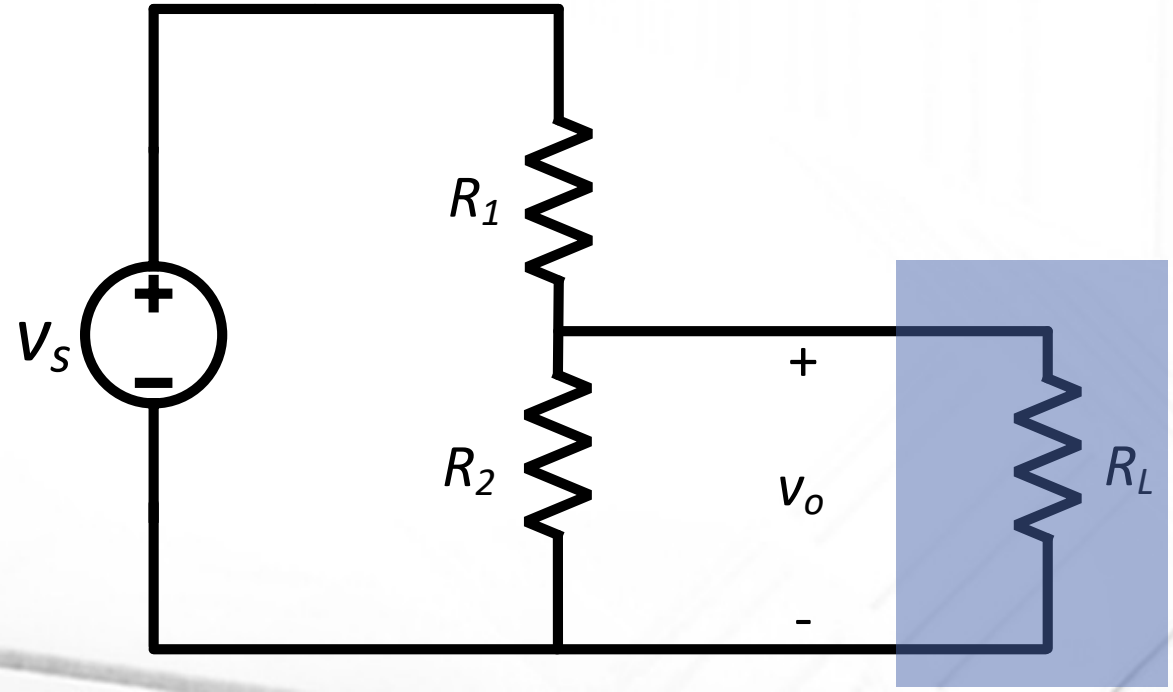
$$v_1 = iR_1 = v_s \frac{R_1}{R_1 + R_2}$$

$$v_2 = iR_2 = v_s \frac{R_2}{R_1 + R_2}$$

Voltage Divider Circuit

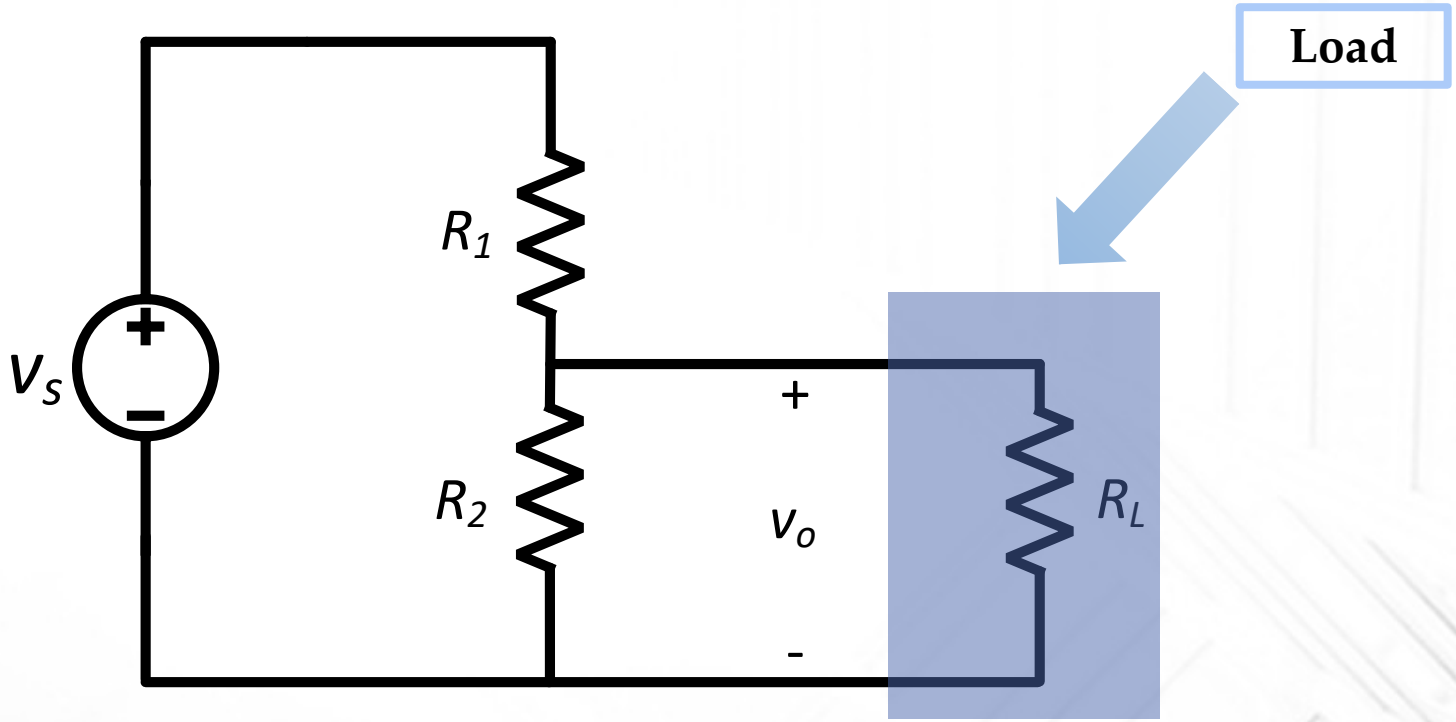
So, by choosing the resistor values appropriately, you can produce the voltage you desire (as long as it's value is less than that of the voltage supply – why?). What if you add another resistor?

This is an example of ***resistor loading*** :



Load

A ***load*** is one or more circuit elements that draw power from the circuit.

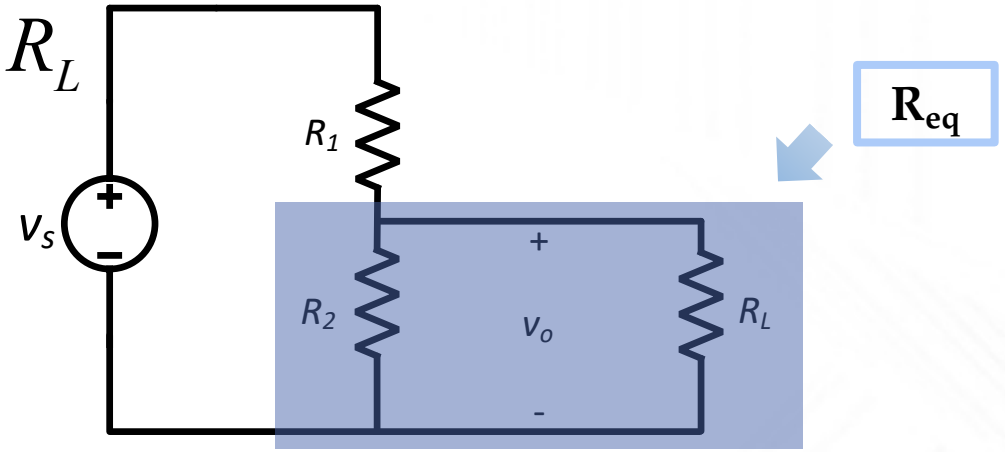


Voltage Divider Circuit: Design

What is the output voltage now?

$$v_o = \frac{R_{eq}}{R_1 + R_{eq}} v_s$$

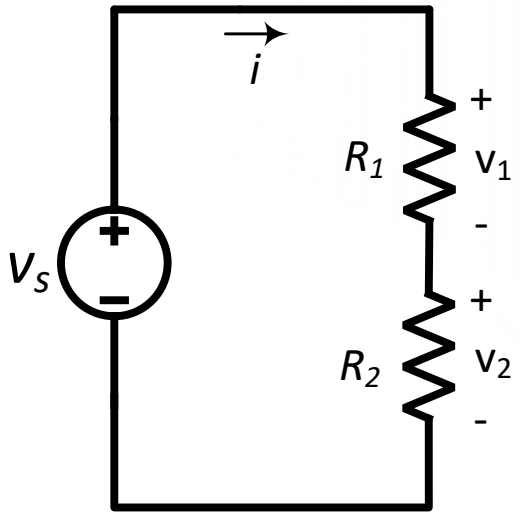
where $R_{eq} = R_2 \parallel R_L$



We find that as long as $R_L \gg R_2$, then the voltage ratio v_o/v_s is not disrupted by the addition of the load! Why?

Voltage Divider Circuit: Design

How do we choose R_1 and R_2 to minimize power losses in the circuit, given that $p_{vs} = {v_s}^2 / {R_1+R_2}$?



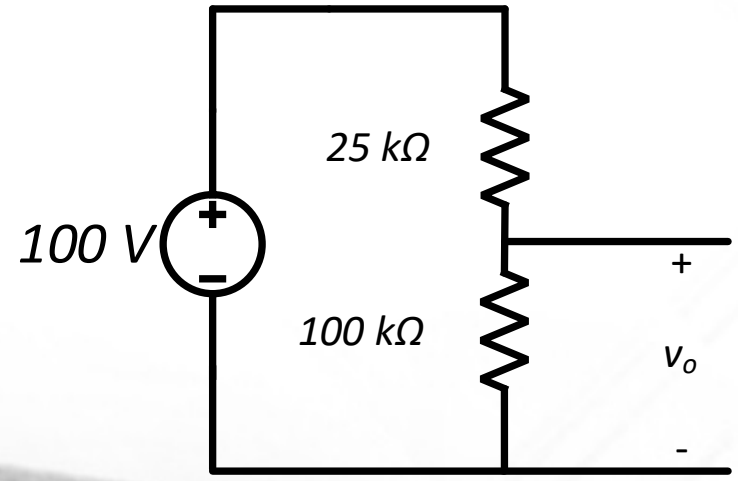
To minimize power losses: $R_1 + R_2 \gg 0$

Voltage Divider Circuit: Design

Another important design consideration is to find out how sensitive the voltage divider circuit is to the tolerance of the resistors...

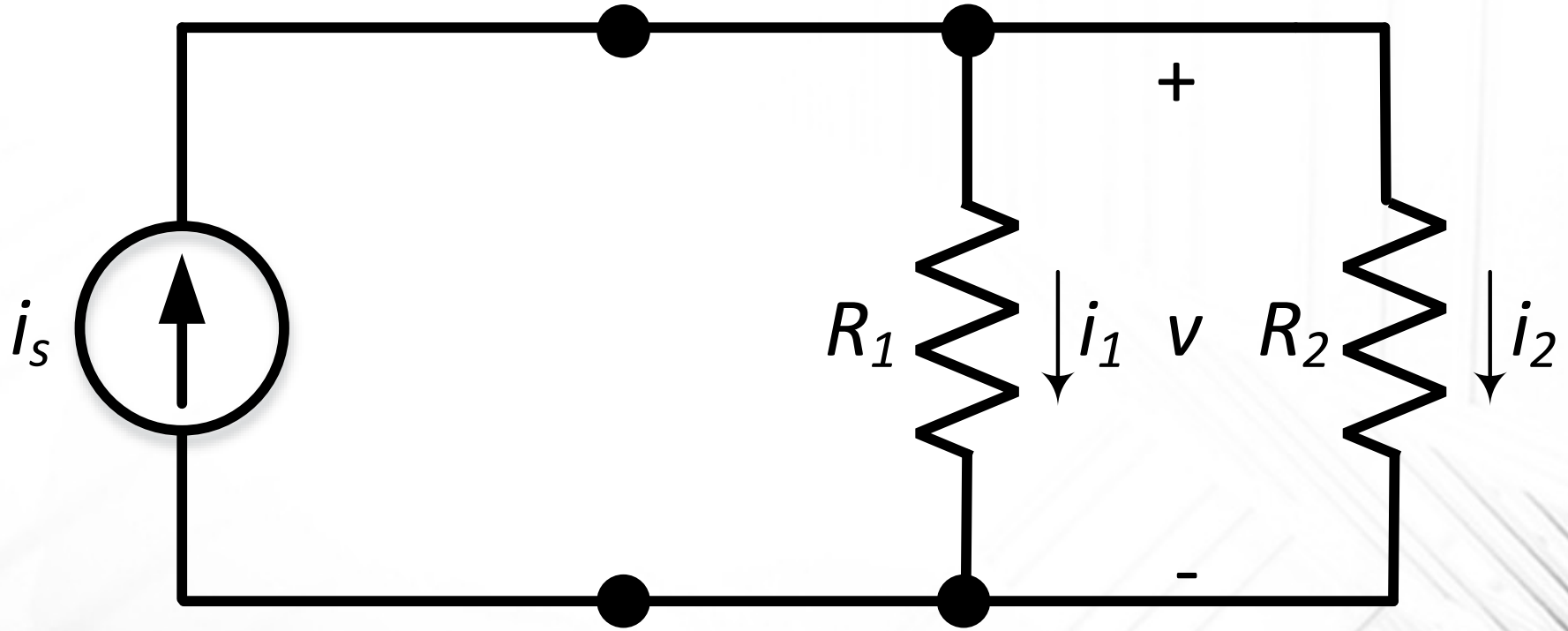
→**Resistor Tolerance**: Commercially available resistors vary within some percentage of their stated value.

Example: Given that the resistors in the voltage divider circuit have a tolerance of +/- 10%, find the maximum and minimum value of v_o.



Current Divider Circuit

We know that when two resistors are in series, the current flowing through them is the same. ***But, what about when two resistors are connected in parallel?***



Current Divider Circuit

The voltage across the parallel resistors is:

$$\boldsymbol{v = i_1 R_1 = i_2 R_2 = \frac{R_1 R_2}{R_1 + R_2} i_s}$$

So, we can find the individual currents, which are:

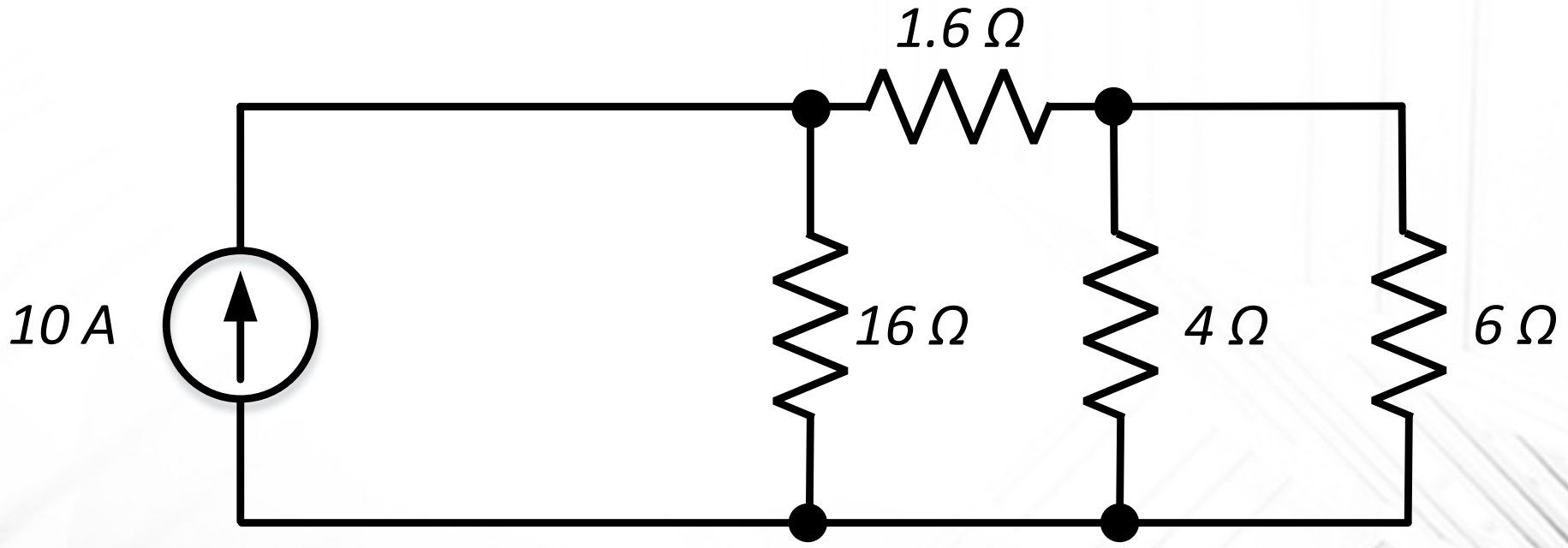
$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

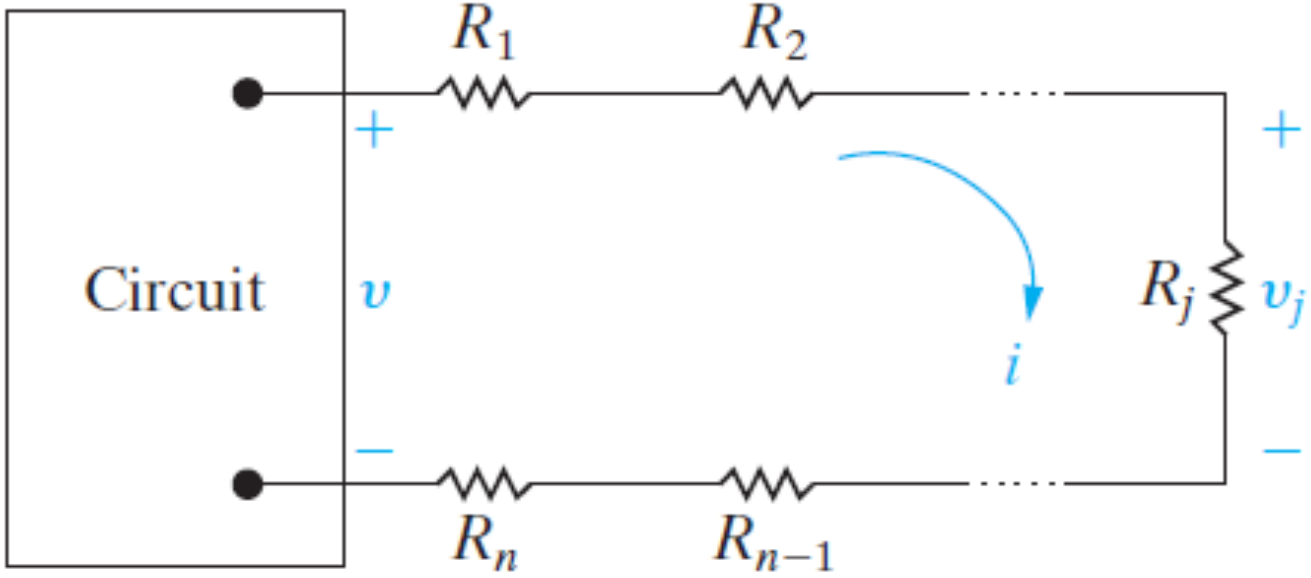
Current Divider Circuit

Let's look at an example, which makes use of the current divider technique.

Example: Find the power dissipated in the 6Ω resistor.

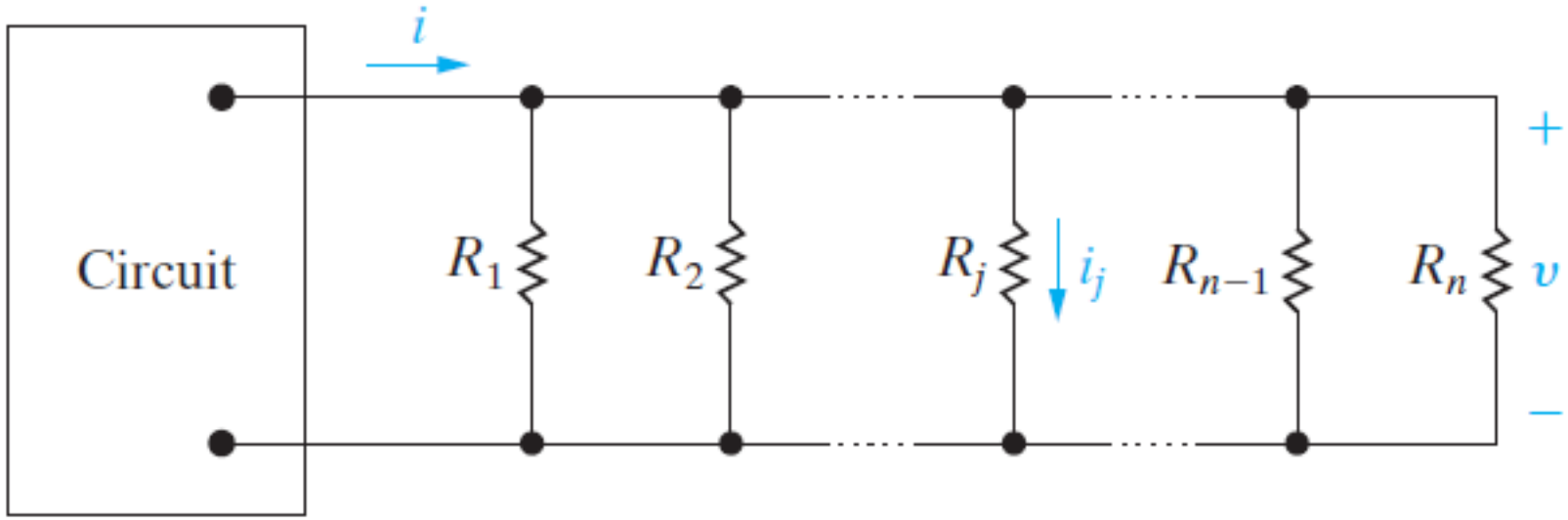


Voltage Division: General Equation



$$v_j = \frac{R_j}{R_{eq}} v$$

Current Division: General Equation



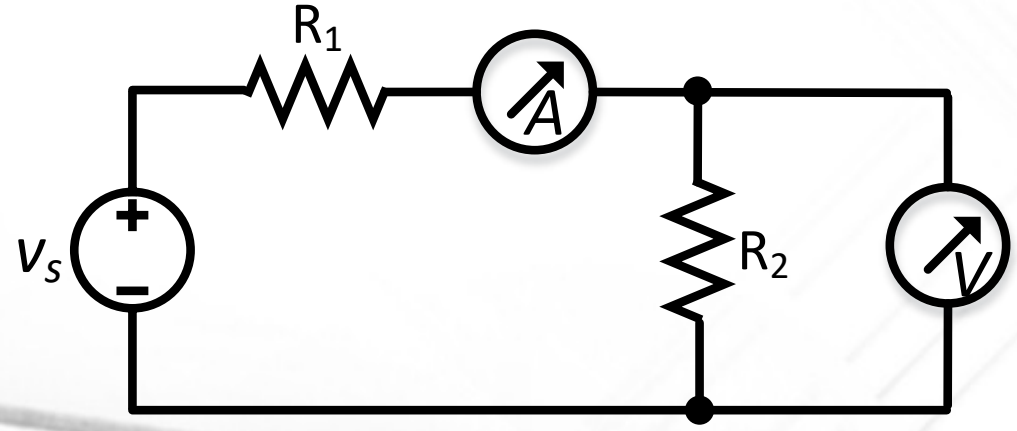
$$i_j = \frac{R_{eq}}{R_j} i$$

Measuring Voltage and Current

When working with circuits, it is often necessary to measure voltages and currents. Here are the tools available at our disposal to measure voltage and current…

Ammeter: This is an instrument that ***measures current***. It must be placed ***in series*** with the circuit element whose current is being measured.

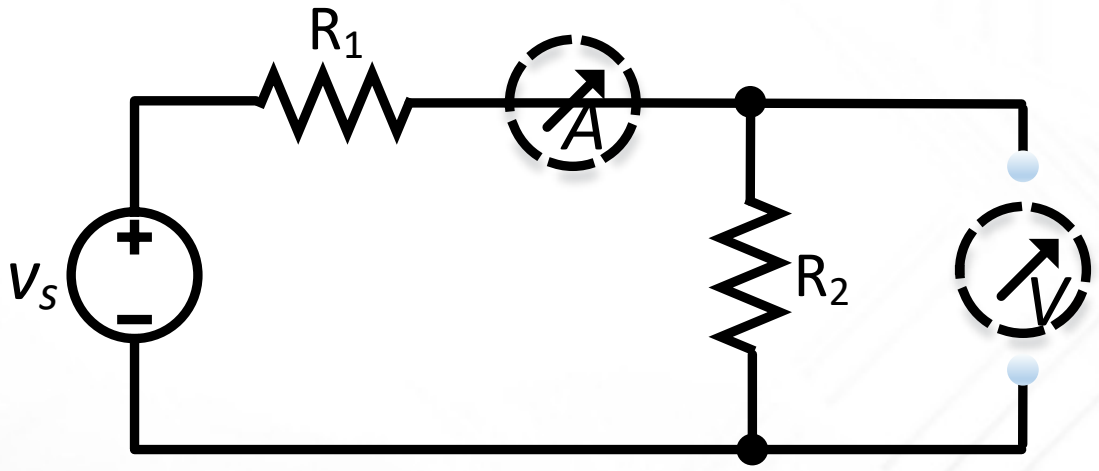
Voltmeter: This is an instrument that ***measures voltage***. It must be placed ***in parallel*** with the circuit element whose voltage is being measured.



Measuring Voltage and Current

Ideal Ammeter: Ideally, an ammeter has no effect on the current being measured. Thus, an ideal ammeter acts as a **short circuit** (equivalent resistance = 0 Ω).

Ideal Voltmeter: Ideally, a voltmeter has no effect on the voltage being measured. Thus, an ideal voltmeter acts as an **open circuit** (equivalent resistance = ∞ Ω).

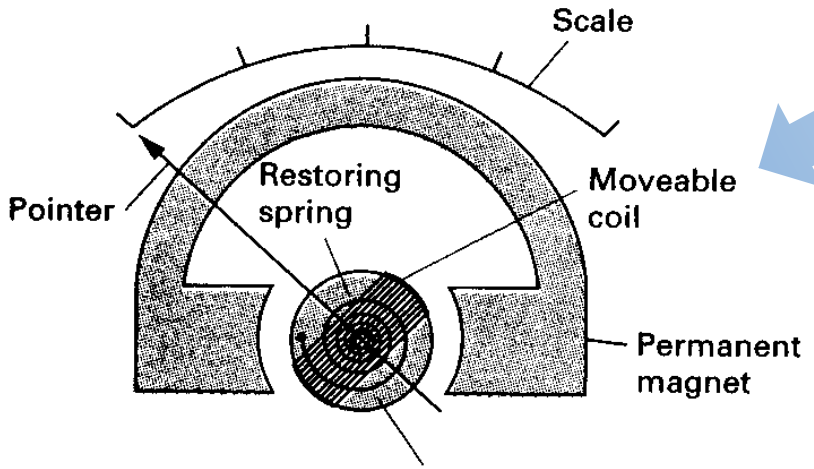


Digital & Analog Meters

Digital Meter: A digital meter measures voltage or current at discrete points in time, called sampling times. They are more commonly seen in the lab, because they are more advantageous than analog meters.

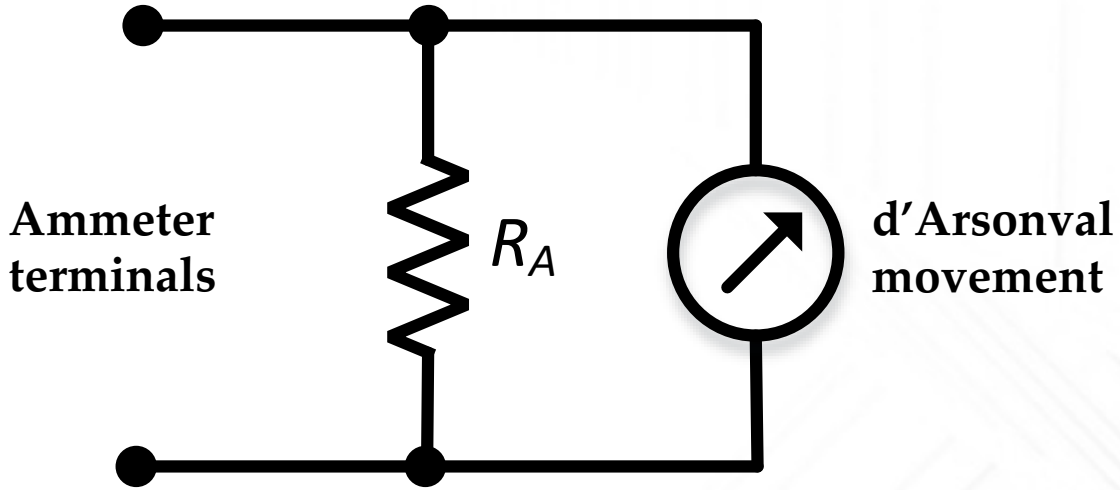
Analog Meter: An analog meter measures voltage or current using d'Arsonval meter movement.

d'Arsonval Meter Movement



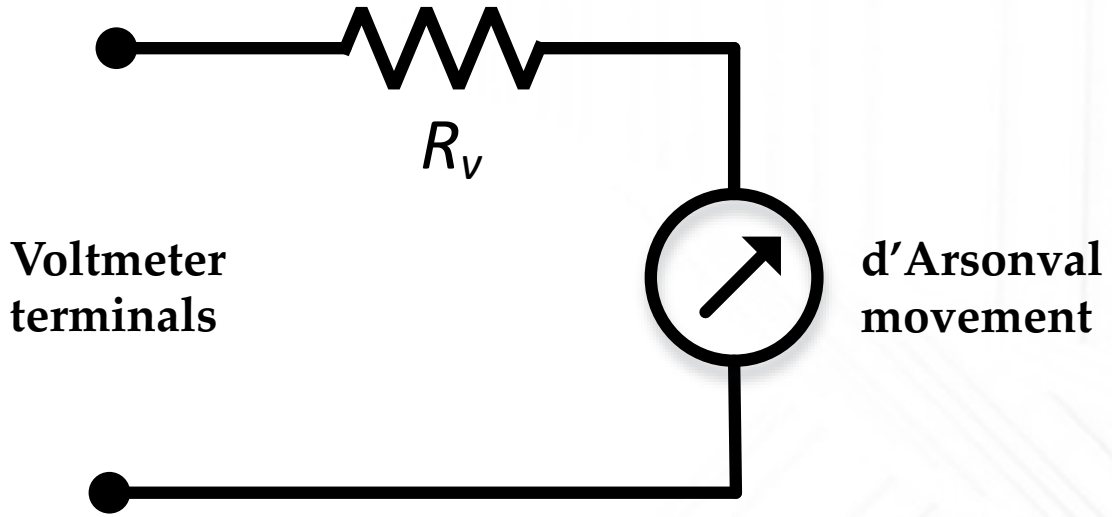
DC Ammeter Circuit

The purpose of the parallel resistor in an analog ammeter is to limit the amount of current in the d'Arsonval meter by shunting some of it through the R_A.



DC Voltmeter Circuit

The purpose of the series resistor in an analog voltmeter is to limit the voltage drop across the d'Arsonval meter.



Designing Meters

We can see that the ammeter and voltmeter are not ideal, because of the presence of additional resistors.

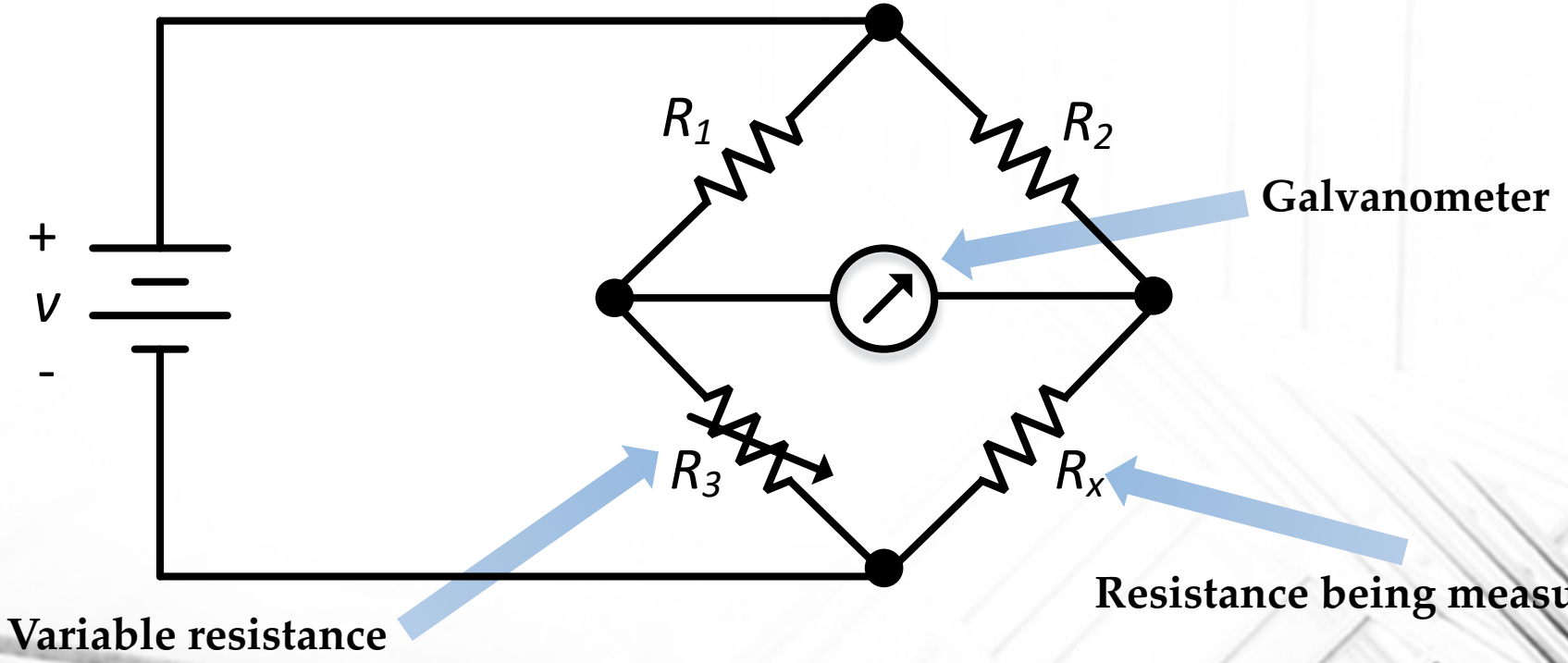
Rule of thumb: When designing an ammeter, its effective resistance should be no more than $1/10^{th}$ of the value of the smallest resistance in the circuit.

Example #1: For a d'Arsonval movement of 50 mV, 1 mA used in an ammeter with a full-scale reading of 10 mA, find R_A.

Example #2: For a d'Arsonval movement of 50 mV, 1 mA used in a voltmeter with a full-scale reading of 150V, find R_v.

Wheatstone Bridge

We've talked about measuring voltage and current, but how do we **measure resistance**? With a wheatstone bridge! (there are other configurations/devices that can also be used to measure resistance)

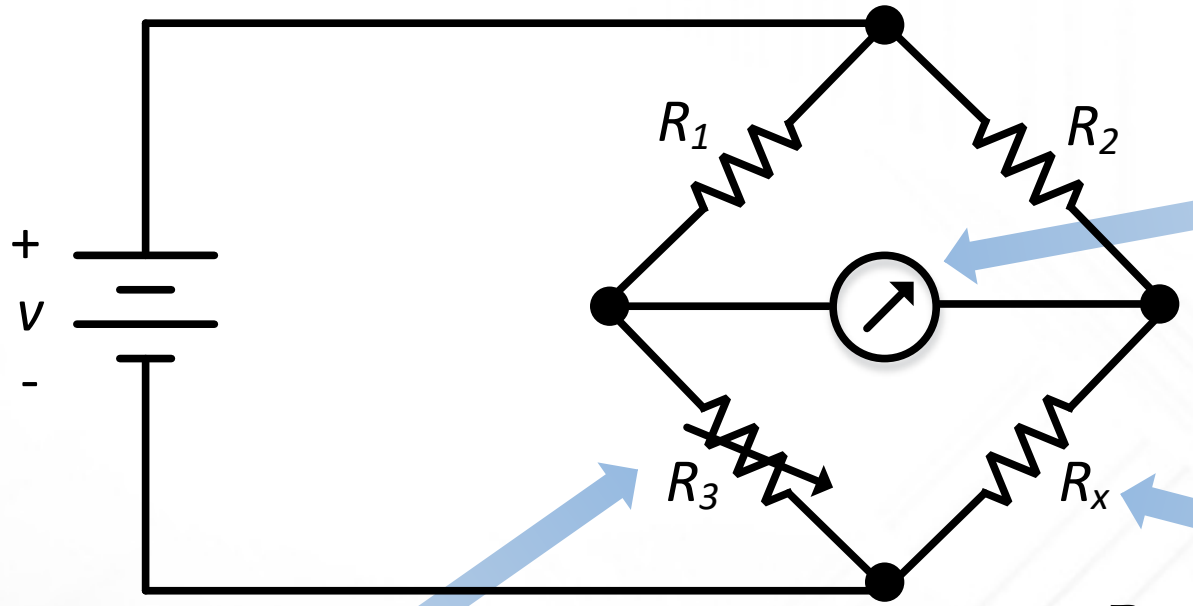


Wheatstone Bridge

Hint: For the current through the galvanometer to be zero, the following must be true:

$$i_1 = i_3$$
$$i_2 = i_x$$

First, here's the idea behind the wheatstone bridge. If there is no current flowing through the galvanometer, that means that $R_x = \frac{R_2}{R_1} R_3$ (**why?**). Since we control R_3 (it is a variable resistor), that means we know its value and hence, the value of R_x.



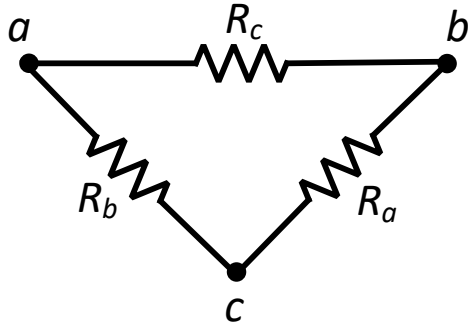
Galvanometer

Variable resistance

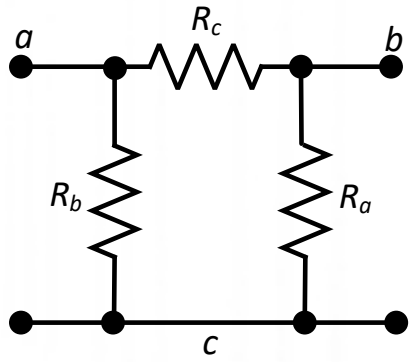
Resistance being measured

Delta-to-Wye Transformation

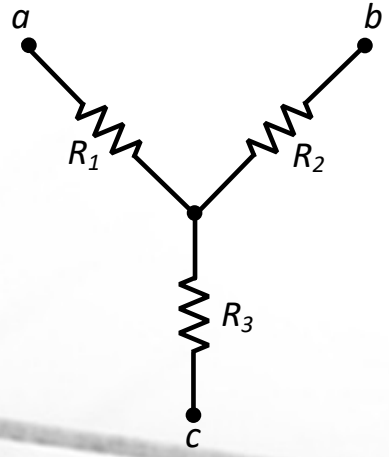
Delta (Δ) Configuration:



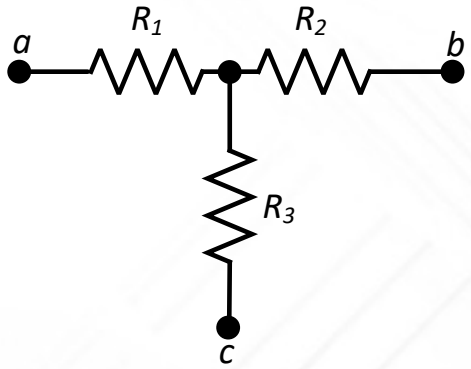
OR



Wye (Y) Configuration:



OR



Delta-to-Wye Transformation

Now, we need to figure out how to get from a delta configuration to a wye configuration (and vice versa)…

$$R_{ab} = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} = R_1 + R_2$$

$$R_{bc} = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} = R_2 + R_3$$

$$R_{ca} = \frac{R_b(R_c + R_a)}{R_a + R_b + R_c} = R_1 + R_3$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

Delta-to-Wye Transformation

Example: Find the current and power supplied by the 40 V source in the circuit shown below.

